



Universidad de San Andrés

Economía Aplicada

Problem Set 6: Difference-in-Differences

Ignacio Anchorena, Rodrigo Braga, Federico Lopez, Joaquin Musich

Fecha de entrega: 28 de Octubre de 2024

1)

La tabla 1 recrea la tabla 4 en Cheng & Hoekstra (2013). En ella se muestran varias regresiones entre crimen y la implementación de doctrina Castle para sugerir que el efecto de disuasión es poco significativo (económicamente). Es decir, tendría sentido enfocarse en el cambio en la violencia neta porque no habría spillover a través del efecto disuasión.

2)

El p-valor del test de tendencias paralelas es 0. Esto estaría invalidando el supuesto de tendencias paralelas antes del tratamiento e invalidaría la estrategia de identificación. En el gráfico 1 podemos ver que, para casi todos los grupos, la tendencia es que antes del tratamiento es muy variable.

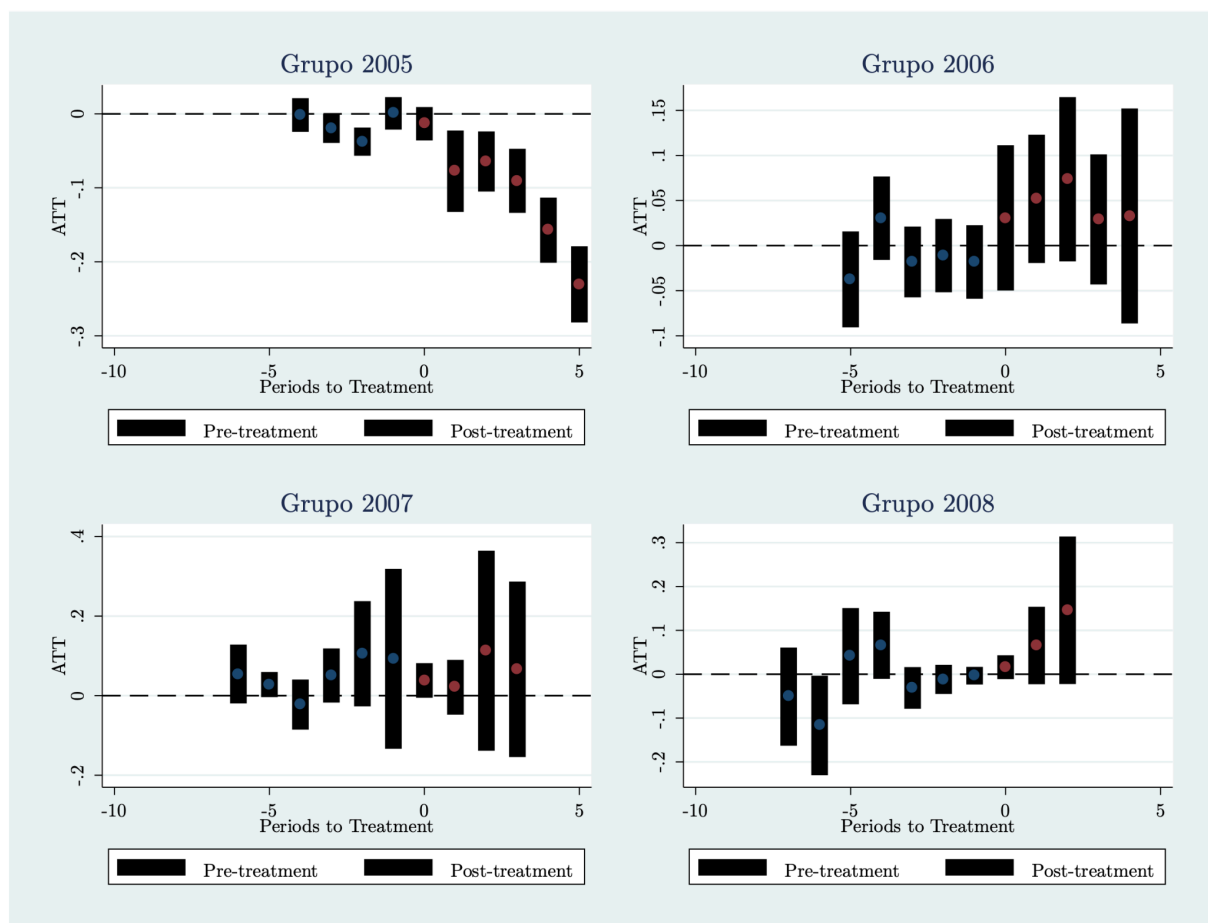


Gráfico 1: Estudio de evento por año de implementación

PANEL A: BURGLARY												
VARIABLES	(1) Log(Burglary Rate)	(2) Log(Burglary Rate)	(3) Log(Burglary Rate)	(4) Log(Burglary Rate)	(5) Log(Burglary Rate)	(6) Log(Burglary Rate)	(7) Log(Burglary Rate)	(8) Log(Burglary Rate)	(9) Log(Burglary Rate)	(10) Log(Burglary Rate)	(11) Log(Burglary Rate)	(12) Log(Burglary Rate)
castle doctrine law, =1 if CDL is effective	0.0780*** (0.0255)	0.0290 (0.0239)	0.0242 (0.0213)	0.0200 (0.0256)	0.0345** (0.0161)	0.0247 (0.0201)	0.0572** (0.0272)	0.0096 (0.0295)	0.0093 (0.0276)	0.0054 (0.0341)	0.0093 (0.0238)	0.0231 (0.0260)
0 to 2 years before adoption of castle doctrine law				-0.0092 (0.0137)						-0.0094 (0.0200)		
PANEL B: ROBBERY												
VARIABLES	(1) Log(Robbery Rate)	(2) Log(Robbery Rate)	(3) Log(Robbery Rate)	(4) Log(Robbery Rate)	(5) Log(Robbery Rate)	(6) Log(Robbery Rate)	(7) Log(Robbery Rate)	(8) Log(Robbery Rate)	(9) Log(Robbery Rate)	(10) Log(Robbery Rate)	(11) Log(Robbery Rate)	(12) Log(Robbery Rate)
castle doctrine law, =1 if CDL is effective	0.0408 (0.0254)	0.0344 (0.0227)	0.0280 (0.0231)	0.0211 (0.0268)	0.0394* (0.0197)	0.0500* (0.0282)	0.0448 (0.0331)	0.0320 (0.0426)	0.0064 (0.0376)	-0.0022 (0.0458)	0.0069 (0.0348)	0.0217 (0.0313)
0 to 2 years before adoption of castle doctrine law				-0.0153 (0.0159)						-0.0205 (0.0248)		
PANEL C: AGGRAVATED ASSAULT												
VARIABLES	(1) Log(Assault Rate)	(2) Log(Assault Rate)	(3) Log(Assault Rate)	(4) Log(Assault Rate)	(5) Log(Assault Rate)	(6) Log(Assault Rate)	(7) Log(Assault Rate)	(8) Log(Assault Rate)	(9) Log(Assault Rate)	(10) Log(Assault Rate)	(11) Log(Assault Rate)	(12) Log(Assault Rate)
castle doctrine law, =1 if CDL is effective	0.0434 (0.0387)	0.0397 (0.0413)	0.0300 (0.0329)	0.0242 (0.0379)	0.0357 (0.0305)	0.0423 (0.0287)	0.0555 (0.0604)	0.0698 (0.0639)	0.0270 (0.0435)	0.0250 (0.0506)	0.0270 (0.0407)	0.0345 (0.0371)
0 to 2 years before adoption of castle doctrine law				-0.0128 (0.0156)						-0.0047 (0.0256)		
Observations	550	550	550	550	550	550	550	550	550	550	550	550
R-squared	0.9629	0.9666	0.9734	0.9734	0.9747	0.9906	0.9462	0.9527	0.9726	0.9726	0.9740	0.9895
State and Year Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region-by-Year Fixed Effects	No	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes
Time-Varying Controls	No	No	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Contemporaneous Crime Rates	No	No	No	No	Yes	No	No	No	No	No	Yes	No
State-Specific Linear Time Trends	No	No	No	No	No	Yes	No	No	No	No	No	Yes

Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates welfare and public assistance spending median income poverty rate unemployment rate and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates.

*Significant at the 10% level**Significant at the 5% level***Significant at the 1% level

Tabla 1: Réplica de la Tabla 4 en Cheng & Hoekstra (2013)

3)

En la tabla 2 vemos el resultado del estimador del efecto de tratamiento usando la metodología de Borusyak et al (2021). Este consiste en usar las variables tratadas en momentos más tardíos para estimar el contrafáctico en el caso de las que fueron tratadas más temprano. El promedio de todas estas diferencias es el tau en la tabla 2. Es un efecto menor al que mencionan Cheng & Hoekstra de 8 o 9 por ciento.

VARIABLES	(1)
	Log(Assault Rate)
tau	0.0395 (0.0409)
Observations	189
Standard errors in parentheses	
*** p<0.01, ** p<0.05, * p<0.1	

Tabla 2: Resultado de imputation estimator de Borusyak et al. (2021)

```
1 // Economía Aplicada Problema set 6
2
3 global main
  "/Users/federicolopez/Library/CloudStorage/OneDrive-Personal/Documents/UDESA/08/APLICADA/TUTORIALES/E178-APLICADA-PS-G1/T06/ProblemSet6_Consignas"
4 global output "$main/output"
5 global input "$main/input"
6 cd "$output"
7
8 ***** Ejercicio 1 *****
9 use "$input/castle.dta", clear
10 *ssc install reghdfe
11 *ssc install ftools
12
13 label var pre2_cdl " 0 to 2 years before adoption of
  castle doctrine law"
14 label var l_burglary "Log(Burglary Rate)"
15 label var l_robbery "Log(Robbery Rate)"
16 label var l_assault "Log(Assault Rate)"
17
18 ***** Panel A Burglary *****
19
20 ** Regresion 1:
21 reghdfe l_burglary cdl [aweight=popwt], absorb(sid year)
  cluster("sid") nocon // el comando [aweight=popwt]
  pondera las observaciones usando la población
22 outreg2 using "$output/Tabla4_Panel_A.rtf", keep(cdl)
  dec(4) label nonot ///
23 addtext(State and Year Fixed Effects, Yes, Region-by-
  Year Fixed Effects, Yes, Time-Varying Controls, Yes,
  Contemporaneous Crime Rates, No, State-Specific Linear
  Time Trends, No) ///
24 addnote (Notes: Each column in each panel represents a
  separate regression. The unit of observation is state-
  year. Robust standard errors are clustered at the state
  level. Time varying controls include policing and
  incarceration rates, welfare and public assistance
  spending, median income, poverty rate, unemployment rate
  , and demographics. Contemporaneous crime rates include
```

```

larceny and motor vehicle theft rates. *Significant at
the 10% level **Significant at the 5% level ***
Significant at the 1% level) replace
25 ** Regresion 2: Vamos a agregar interacciones por año y
región.
26 gen northeast_year = northeast*year
27 gen south_year = south*year
28 gen midwest_year = midwest*year
29 gen west_year = west*year
30
31 reghdfe l_burglary cdl [aweight=popwt], absorb(sid year
northeast_year south_year midwest_year west_year)
cluster("sid") nocon
32 outreg2 using "$output/Tabla4_Panel_A.rtf", dec(4) label
nonot addtext(State and Year Fixed Effects, Yes, Region
-by-Year Fixed Effects, Yes, Time-Varying Controls, No,
Contemporaneous Crime Rates, No, State-Specific Linear
Time Trends, No) ///
33 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level **Significant at the 5% level ***Significant
at the 1% level) append
34
35 ** Regresion 3: Vamos a agregar controles que varían
con el tiempo.
36 reghdfe l_burglary cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44
whitem_15_24 whitem_15_24 [aweight=popwt], absorb(sid
year northeast_year south_year midwest_year west_year)
cluster("sid") nocon
37 outreg2 using "$output/Tabla4_Panel_A.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-

```

Specific Linear Time Trends, No) ///

38 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

39

40 ** Regresion 4: Ahora incluimos una variable adicional
que indica sobre los dos años antes de implementar la
ley.

41 reghdfe l_burglary cdl pre2_cdl l_police prisoner
exp_pubwelfare income poverty unemployrt blackm_15_24
blackm_25_44 whitem_15_24 whitem_15_24 [aweight=popwt],
absorb(sid year northeast_year south_year midwest_year
west_year) cluster("sid") nocon
42 outreg2 using "\$output/Tabla4_Panel_A.rtf", keep(cdl
pre2_cdl) dec(4) label nonot addtext(State and Year
Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes,
Time-Varying Controls, Yes, Contemporaneous Crime Rates,
No, State-Specific Linear Time Trends, No) ///

43 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

44

45 ** Regresion 5: Ahora vamos a agregar variables de
control. Logaritmo de robo contemporaneo, robo de
vehículo de motor y descartamos la variable sobre los
dos años anteriores a la implementacion de la ley.

```

46  reghdfe l_burglary cdl l_larceny l_motor l_police
    prisoner exp_pubwelfare income poverty unemployrt
    blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24 [
    aweight=popwt], absorb(sid year northeast_year
    south_year midwest_year west_year) cluster("sid") nocon
47  outreg2 using "$output/Tabla4_Panel_A.rtf", keep(cdl)
    dec(4) label nonot addtext(State and Year Fixed Effects,
    Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
    Controls, Yes, Contemporaneous Crime Rates, Yes, State-
    Specific Linear Time Trends, No) ///
48  addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
    larceny and motor vehicle theft rates. *Significant at
    the 10% level**Significant at the 5% level***Signifi
    cant at the 1% level) append
49
50  ** Regresion 6: Y finalmente agregamos las tendencias
    lineales de cada Estado.
51  reghdfe l_burglary cdl l_police prisoner exp_pubwelfare
    income poverty unemployrt blackm_15_24 blackm_25_44
    whitem_15_24 whitem_15_24 trend_1-trend_51 [aweight=
    popwt], absorb(sid year northeast_year south_year
    midwest_year west_year) cluster("sid") nocon
52  outreg2 using "$output/Tabla4_Panel_A.rtf", keep(cdl)
    dec(4) label nonot addtext(State and Year Fixed Effects,
    Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
    Controls, Yes, Contemporaneous Crime Rates, No, State-
    Specific Linear Time Trends, Yes) ///
53  addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include

```

larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

54

55 // Ahora en las regresiones 7 a 12 vamos a usar el mismo procedimiento pero sin ponderar por la población de cada Estado.

56

57 ** Regresion 7:

58 reghdfe l_burglary cdl, absorb(sid year) cluster("sid") nocon

59 outreg2 using "\$output/Tabla4_Panel_A.rtf", dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region -by-Year Fixed Effects, No, Time-Varying Controls, No, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

60 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

61

62 ** Regresion 8: Vamos a agregar interacciones por año y región.

63 reghdfe l_burglary cdl, absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon
64 outreg2 using "\$output/Tabla4_Panel_A.rtf", dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region -by-Year Fixed Effects, Yes, Time-Varying Controls, No, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

65 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and

incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

66

67 ** Regresion 9: Vamos a agregar controles que varían con el tiempo.

68 reghdfe l_burglary cdl l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

69 outreg2 using "\$output/Tabla4_Panel_A.rtf", keep(cdl) dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

70 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

71

72 ** Regresion 10: Ahora incluimos una variable adicional que indica sobre los dos años antes de implementar la ley.

73 reghdfe l_burglary cdl pre2_cdl l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

74 outreg2 using "\$output/Tabla4_Panel_A.rtf", keep(cdl pre2_cdl) dec(4) label nonot addtext(State and Year

Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

75 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

76

77 ** Regresion 11: Ahora vamos a agregar variables de control. Logaritmo de robo contemporaneo, robo de vehículo de motor y descartamos la variable sobre los dos años anteriores a la implementacion de la ley.

78 reghdfe l_burglary cdl l_larceny l_motor l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

79 outreg2 using "\$output/Tabla4_Panel_A.rtf", keep(cdl) dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, Yes, State-Specific Linear Time Trends, No) ///

80 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

81

82 ** Regresion 12: Y finalmente agregamos las tendencias

lineales de cada Estado.

```

83 reghdfe l_burglary cdl l_police prisoner exp_pubwelfare
    income poverty unemployrt blackm_15_24 blackm_25_44
    whitem_15_24 whitem_15_24 trend_1-trend_51, absorb(sid
    year northeast_year south_year midwest_year west_year)
    cluster("sid") nocon
84 outreg2 using "$output/Tabla4_Panel_A.rtf", keep(cdl)
    dec(4) label nonot addtext(State and Year Fixed Effects,
    Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
    Controls, Yes, Contemporaneous Crime Rates, No, State-
    Specific Linear Time Trends, Yes) ///
85 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
    larceny and motor vehicle theft rates. *Significant at
    the 10% level**Significant at the 5% level***Significant
    at the 1% level) append
86
87 ***** Panel B Robbery *****
88
89 * Lo que vamos a hacer acá es lo mismo que en el Panel
    A, pero con l_robbery como la variable dependiente en
    vez de l_burglary
90 ** Regresion 1:
91 reghdfe l_robbery cdl [aweight=popwt], absorb(sid year)
    cluster("sid") nocon // el comando [aweight=popwt]
    pondera las observaciones usando la población
92 outreg2 using "$output/Tabla4_Panel_B.rtf", dec(4) label
    nonot addtext(State and Year Fixed Effects, Yes, Region
    -by-Year Fixed Effects, No, Time-Varying Controls, No,
    Contemporaneous Crime Rates, No, State-Specific Linear
    Time Trends, No) ///
93 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and

```

incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate , and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) replace

94

95 ** Regresion 2: Vamos a agregar interacciones por año y región.

96 reghdfe l_robbery cdl [aweight=popwt], absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

97 outreg2 using "\$output/Tabla4_Panel_B.rtf", dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region -by-Year Fixed Effects, Yes, Time-Varying Controls, No, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

98 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate , and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

99

100 ** Regresion 3: Vamos a agregar controles que varían con el tiempo.

101 reghdfe l_robbery cdl l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24 [aweight=popwt], absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

102 outreg2 using "\$output/Tabla4_Panel_B.rtf", keep(cdl) dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

```

103 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
    larceny and motor vehicle theft rates. *Significant at
    the 10% level**Significant at the 5% level***Significant
    at the 1% level) append

104
105 ** Regresion 4: Ahora incluimos una variable adicional
    que indica sobre los dos años antes de implementar la
    ley.
106 reghdfe l_robbery cdl pre2_cdl l_police prisoner
    exp_pubwelfare income poverty unemployrt blackm_15_24
    blackm_25_44 whitem_15_24 whitem_15_24 [aweight=popwt],
    absorb(sid year northeast_year south_year midwest_year
    west_year) cluster("sid") nocon
107 outreg2 using "$output/Tabla4_Panel_B.rtf", keep(cdl
    pre2_cdl) dec(4) label nonot addtext(State and Year
    Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes,
    Time-Varying Controls, Yes, Contemporaneous Crime Rates,
    No,State-Specific Linear Time Trends, No) ///
108 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
    larceny and motor vehicle theft rates. *Significant at
    the 10% level**Significant at the 5% level***Significant
    at the 1% level) append

109
110 ** Regresion 5: Ahora vamos a agregar variables de
    control. Logaritmo de robo contemporaneo, robo de
    vehículo de motor y descartamos la variable sobre los
    dos años anteriores a la implementacion de la ley.
111 reghdfe l_robbery cdl l_larceny l_motor l_police

```

```

prisoner exp_pubwelfare income poverty unemployrt
blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24 [
aweight=popwt], absorb(sid year northeast_year
south_year midwest_year west_year) cluster("sid") nocon
112 outreg2 using "$output/Tabla4_Panel_B.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, Yes, State-
Specific Linear Time Trends, No) ///
113 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Signifi-
cant at the 1% level) append
114
115 ** Regresion 6: Y finalmente agregamos las tendencias
lineales de cada Estado.
116 reghdfe l_robbery cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44
whitem_15_24 whitem_15_24 trend_1-trend_51 [aweight=
popwt], absorb(sid year northeast_year south_year
midwest_year west_year) cluster("sid") nocon
117 outreg2 using "$output/Tabla4_Panel_B.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-
Specific Linear Time Trends, Yes) ///
118 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at

```

the 10% level**Significant at the 5% level***Significant
at the 1% level) append

119

120 // Ahora en las regresiones 7 a 12 vamos a usar el
mismo procedimiento pero sin ponderar por la población
de cada Estado.

121

122 ** Regresion 7:

123 reghdfe l_robbery cdl, absorb(sid year) cluster("sid")
nocon

124 outreg2 using "\$output/Tabla4_Panel_B.rtf", dec(4) label
nonot addtext(State and Year Fixed Effects, Yes, Region
-by-Year Fixed Effects, No, Time-Varying Controls, No,
Contemporaneous Crime Rates, No, State-Specific Linear
Time Trends, No) ///

125 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

126

127 ** Regresion 8: Vamos a agregar interacciones por año y
región.

128 reghdfe l_robbery cdl, absorb(sid year northeast_year
south_year midwest_year west_year) cluster("sid") nocon
129 outreg2 using "\$output/Tabla4_Panel_B.rtf", dec(4) label
nonot addtext(State and Year Fixed Effects, Yes, Region
-by-Year Fixed Effects, Yes, Time-Varying Controls, No,
Contemporaneous Crime Rates, No, State-Specific Linear
Time Trends, No) ///

130 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance

spending, median income, poverty rate, unemployment rate
 , and demographics. Contemporaneous crime rates include
 larceny and motor vehicle theft rates. *Significant at
 the 10% level**Significant at the 5% level***Significant
 at the 1% level) append

131

132 ** Regresion 9: Vamos a agregar controles que varían
 con el tiempo.

133 reghdfe l_robbery cdl l_police prisoner exp_pubwelfare
 income poverty unemployrt blackm_15_24 blackm_25_44
 whitem_15_24 whitem_15_24, absorb(sid year
 northeast_year south_year midwest_year west_year)
 cluster("sid") nocon

134 outreg2 using "\$output/Tabla4_Panel_B.rtf", keep(cdl)
 dec(4) label nonot addtext(State and Year Fixed Effects,
 Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
 Controls, Yes, Contemporaneous Crime Rates, No, State-
 Specific Linear Time Trends, No) ///

135 addnote (Notes: Each column in each panel represents a
 separate regression. The unit of observation is state-
 year. Robust standard errors are clustered at the state
 level. Time varying controls include policing and
 incarceration rates, welfare and public assistance
 spending, median income, poverty rate, unemployment rate
 , and demographics. Contemporaneous crime rates include
 larceny and motor vehicle theft rates. *Significant at
 the 10% level**Significant at the 5% level***Significant
 at the 1% level) append

136

137 ** Regresion 10: Ahora incluimos una variable adicional
 que indica sobre los dos años antes de implementar la
 ley.

138 reghdfe l_robbery cdl pre2_cdl l_police prisoner
 exp_pubwelfare income poverty unemployrt blackm_15_24
 blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year
 northeast_year south_year midwest_year west_year)
 cluster("sid") nocon

139 outreg2 using "\$output/Tabla4_Panel_B.rtf", keep(cdl
 pre2_cdl) dec(4) label nonot addtext(State and Year
 Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes,

Time-Varying Controls, Yes, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

140 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

141

142 ** Regresion 11: Ahora vamos a agregar variables de control. Logaritmo de robo contemporaneo, robo de vehículo de motor y descartamos la variable sobre los dos años anteriores a la implementacion de la ley.

143 reghdfe l_robbery cdl l_larceny l_motor l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

144 outreg2 using "\$output/Tabla4_Panel_B.rtf", keep(cdl) dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, Yes, State-Specific Linear Time Trends, No) ///

145 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

146

147 ** Regresion 12: Y finalmente agregamos las tendencias lineales de cada Estado.

```

148 reghdfe l_robbery cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44
whitem_15_24 whitem_15_24 trend_1-trend_51, absorb(sid
year northeast_year south_year midwest_year west_year)
cluster("sid") nocon
149 outreg2 using "$output/Tabla4_Panel_B.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-
Specific Linear Time Trends, Yes) ///
150 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append
151
152 ***** Panel C Assault *****
153
154 * Vamos a hacer lo mismo que con las otras dos
variables dependientes pero con l_assault.
155 ** Regresion 1:
156 reghdfe l_assault cdl [aweight=popwt], absorb(sid year)
cluster("sid") nocon // el comando [aweight=popwt]
pondera las observaciones usando la población
157 outreg2 using "$output/Tabla4_Panel_C.rtf", dec(4) label
nonot addtext(State and Year Fixed Effects, Yes, Region
-by-Year Fixed Effects, No, Time-Varying Controls, No,
Contemporaneous Crime Rates, No, State-Specific Linear
Time Trends, No) ///
158 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate

```

, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) replace

159

160 ** Regresion 2: Vamos a agregar interacciones por año y región.

161 reghdfe l_assault cdl [aweight=popwt], absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

162 outreg2 using "\$output/Tabla4_Panel_C.rtf", dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, No, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

163 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

164

165 ** Regresion 3: Vamos a agregar controles que varían con el tiempo.

166 reghdfe l_assault cdl l_police prisoner exp_pubwelfare income poverty unemployrt blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24 [aweight=popwt], absorb(sid year northeast_year south_year midwest_year west_year) cluster("sid") nocon

167 outreg2 using "\$output/Tabla4_Panel_C.rtf", keep(cdl) dec(4) label nonot addtext(State and Year Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes, Time-Varying Controls, Yes, Contemporaneous Crime Rates, No, State-Specific Linear Time Trends, No) ///

168 addnote (Notes: Each column in each panel represents a separate regression. The unit of observation is state-

year. Robust standard errors are clustered at the state level. Time varying controls include policing and incarceration rates, welfare and public assistance spending, median income, poverty rate, unemployment rate, and demographics. Contemporaneous crime rates include larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

169

170 ** Regresion 4: Ahora incluimos una variable adicional que indica sobre los dos años antes de implementar la ley.

171 reghdfe l_assault cdl pre2_cdl l_police prisoner
exp_pubwelfare income poverty unemployrt blackm_15_24
blackm_25_44 whitem_15_24 whitem_15_24 [aweight=popwt],
absorb(sid year northeast_year south_year midwest_year
west_year) cluster("sid") nocon

172 outreg2 using "\$output/Tabla4_Panel_C.rtf", keep(cdl
pre2_cdl) dec(4) label nonot addtext(State and Year
Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes,
Time-Varying Controls, Yes, Contemporaneous Crime Rates,
No, State-Specific Linear Time Trends, No) ///

173 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

174

175 ** Regresion 5: Ahora vamos a agregar variables de
control. Logaritmo de robo contemporaneo, robo de
vehículo de motor y descartamos la variable sobre los
dos años anteriores a la implementacion de la ley.

176 reghdfe l_assault cdl l_larceny l_motor l_police
prisoner exp_pubwelfare income poverty unemployrt
blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24 [

```

177   aweight=popwt], absorb(sid year northeast_year
south_year midwest_year west_year) cluster("sid") nocon
outreg2 using "$output/Tabla4_Panel_C.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, Yes, State-
Specific Linear Time Trends, No) ///
178   addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Signifi-
cant at the 1% level) append
179
180   ** Regresion 6: Y finalmente agregamos las tendencias
lineales de cada Estado.
181   reghdfe l_assault cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44
whitem_15_24 whitem_15_24 trend_1-trend_51 [aweight=
popwt], absorb(sid year northeast_year south_year
midwest_year west_year) cluster("sid") nocon
182   outreg2 using "$output/Tabla4_Panel_C.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-
Specific Linear Time Trends, Yes) ///
183   addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

```

```
184
185 // Ahora en las regresiones 7 a 12 vamos a usar el
    mismo procedimiento pero sin ponderar por la población
    de cada Estado.
186
187 ** Regresion 7:
188 reghdfe l_assault cdl, absorb(sid year) cluster("sid")
    nocon
189 outreg2 using "$output/Tabla4_Panel_C.rtf", dec(4) label
    nonot addtext(State and Year Fixed Effects, Yes, Region
    -by-Year Fixed Effects, No, Time-Varying Controls, No,
    Contemporaneous Crime Rates, No, State-Specific Linear
    Time Trends, No) ///
190 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
    larceny and motor vehicle theft rates. *Significant at
    the 10% level**Significant at the 5% level***Significant
    at the 1% level) append
191
192 ** Regresion 8: Vamos a agregar interacciones por año y
    región.
193 reghdfe l_assault cdl, absorb(sid year northeast_year
    south_year midwest_year west_year) cluster("sid") nocon
194 outreg2 using "$output/Tabla4_Panel_C.rtf", dec(4) label
    nonot addtext(State and Year Fixed Effects, Yes, Region
    -by-Year Fixed Effects, Yes, Time-Varying Controls, No,
    Contemporaneous Crime Rates, No, State-Specific Linear
    Time Trends, No) ///
195 addnote (Notes: Each column in each panel represents a
    separate regression. The unit of observation is state-
    year. Robust standard errors are clustered at the state
    level. Time varying controls include policing and
    incarceration rates, welfare and public assistance
    spending, median income, poverty rate, unemployment rate
    , and demographics. Contemporaneous crime rates include
```

larceny and motor vehicle theft rates. *Significant at the 10% level**Significant at the 5% level***Significant at the 1% level) append

196

197 ** Regresion 9: Vamos a agregar controles que varían con el tiempo.

198 reghdfe l_assault cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44
whitem_15_24 whitem_15_24, absorb(sid year
northeast_year south_year midwest_year west_year)
cluster("sid") nocon

199 outreg2 using "\$output/Tabla4_Panel_C.rtf", keep(cdl
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-
Specific Linear Time Trends, No) ///

200 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append

201

202 ** Regresion 10: Ahora incluimos una variable adicional
que indica sobre los dos años antes de implementar la
ley.

203 reghdfe l_assault cdl pre2_cdl l_police prisoner
exp_pubwelfare income poverty unemployrt blackm_15_24
blackm_25_44 whitem_15_24 whitem_15_24, absorb(sid year
northeast_year south_year midwest_year west_year)
cluster("sid") nocon

204 outreg2 using "\$output/Tabla4_Panel_C.rtf", keep(cdl
pre2_cdl) dec(4) label nonot addtext(State and Year
Fixed Effects, Yes, Region-by-Year Fixed Effects, Yes,
Time-Varying Controls, Yes, Contemporaneous Crime Rates,
No, State-Specific Linear Time Trends, No) ///

205 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Signifi-
cant at the 1% level) append

206

207 ** Regresion 11: Ahora vamos a agregar variables de
control. Logaritmo de robo contemporaneo, robo de
vehículo de motor y descartamos la variable sobre los
dos años anteriores a la implementacion de la ley.
208 reghdfe l_assault cdl l_larceny l_motor l_police
prisoner exp_pubwelfare income poverty unemployrt
blackm_15_24 blackm_25_44 whitem_15_24 whitem_15_24,
absorb(sid year northeast_year south_year midwest_year
west_year) cluster("sid") nocon
209 outreg2 using "\$output/Tabla4_Panel_C.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, Yes, State-
Specific Linear Time Trends, No) ///

210 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Signifi-
cant at the 1% level) append

211

212 ** Regresion 12: Y finalmente agregamos las tendencias
lineales de cada Estado.

213 reghdfe l_assault cdl l_police prisoner exp_pubwelfare
income poverty unemployrt blackm_15_24 blackm_25_44

```

whitem_15_24 whitem_15_24 trend_1-trend_51, absorb(sid
year northeast_year south_year midwest_year west_year)
cluster("sid") nocon
214 outreg2 using "$output/Tabla4_Panel_C.rtf", keep(cdl)
dec(4) label nonot addtext(State and Year Fixed Effects,
Yes, Region-by-Year Fixed Effects, Yes, Time-Varying
Controls, Yes, Contemporaneous Crime Rates, No, State-
Specific Linear Time Trends, Yes) ///
215 addnote (Notes: Each column in each panel represents a
separate regression. The unit of observation is state-
year. Robust standard errors are clustered at the state
level. Time varying controls include policing and
incarceration rates, welfare and public assistance
spending, median income, poverty rate, unemployment rate
, and demographics. Contemporaneous crime rates include
larceny and motor vehicle theft rates. *Significant at
the 10% level**Significant at the 5% level***Significant
at the 1% level) append
216
217 ***** Ejercicio 2 *****
218
219 ** Instalamos los paquetes necesarios para usar
Callaway & Sant'Anna (ya instalados los paquetes los
ponemos en formato de texto para que corra mas rapido
el codigo)
220 *ssc install drdid
221 *ssc install csdid
222
223 ** Para los años donde no fue tratado necesitamos que
haya un 0 en vez de missing value "."
224 replace effyear = 0 if effyear = .
225
226 ** Callaway & Sant'Anna
227 csdid l_assault cdl, ivar(sid) time(year) gvar(effyear)
method(reg) notyet
228
229 ** ATT simple
230 estat simple
231
232 ** Gráfico

```

```
233 csdid l_assault cdl, ivar(sid) time(year) gvar(effyear)
    method(reg) notyet
234 csdid_plot, group(2005) name(m1,replace) title("Grupo
    2005")
235 csdid_plot, group(2006) name(m2,replace) title("Grupo
    2006")
236 csdid_plot, group(2007) name(m3,replace) title("Grupo
    2007")
237 csdid_plot, group(2008) name(m4,replace) title("Grupo
    2008")
238 graph combine m1 m2 m3 m4, xcommon scale(0.8)
239
240 graph save "$output/Graph_EJ2.gph", replace
241
242 ** P-valor del test
243 estat pretrend
244
245 ***** Ejercicio 3 *****
246
247 ** Imputation estimation de Borusyak et al.(2021).
248 ** Instalamos los paquetes necesarios para usar
    Borusyak (ya instalados los paquetes los ponemos en
    formato de texto para que corra mas rapido el codigo)
249
250 *ssc install did_imputation
251 did_imputation l_assault sid year effyear, autosample
252
253 outreg2 using "$output/did_imputation.rtf", replace
254
255
256
257
258
259
260
261
262
263
264
265
```

266
267
268